Solution Architecture

Date	
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

PRESENTATION LAYER

▼

Web Browser (Desktop / Tablet / Mobile)
HTML5 · CSS3 · Bootstrap 5 · Chart.js · Jinja2 Templates

APPLICATION LAYER

▼

Flask Web Server (Python 3.14)
├── Routes: /, /recommend, /suitability, /dashboard, /about, /contact
├── ML Prediction Engine (Random Forest 99.5% accuracy)
├── Suitability Assessment Module
├── REST API (/api/predict, /api/metrics)
└── K-Means Exploratory Clustering

DATA LAYER

Crop_recommendation.csv (2200 rows, 22 crops, 7 features)
├── crop_model.pkl (trained Random Forest)
├── label_encoder.pkl (LabelEncoder)
├── metrics.json (model comparison results)
└── pca_data.pkl (K-Means cluster data for visualization)

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the user interface for crop recommendation, suitability analysis, and research dashboard. Handles user input forms, displays prediction results with confidence scores, presents EDA and model comparison charts, and provides navigation across all pages.	HTML5, CSS3, Bootstrap 5, Chart.js, Jinja2 Templates
Flask Web Server (API Gateway / Controller)	Acts as the central request handler: routes HTTP requests, validates input, invokes the ML prediction engine, and returns rendered templates or JSON responses. Serves static assets (images, CSS, JS) and exposes REST endpoints for external integration.	Flask 3.1, Werkzeug, Jinja2
ML Prediction Engine (Application Core)	Core business logic: loads the trained Random Forest model and label encoder, constructs a feature DataFrame from user input, predicts the optimal crop, computes confidence from predict_proba, and executes the full suitability assessment with environmental analysis.	scikit-learn 1.9 (RandomForestClassifier), pandas 3.x, numpy 2.x, joblib
Data Store	Persists all project data: the CSV dataset used for training, serialized model artifacts (model, encoder, PCA data), and evaluation metrics in JSON format. All files reside on the local filesystem within the project directory.	CSV (.csv), Pickle (.pkl), JSON (.json), Local Filesystem